

Notes: Ahern & Dittmar (QJE, 2012)

The Changing of the Boards: The Impact on Firm Valuation of Mandated Female Board Representation

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1 Big picture

- **Question.** What happens to firm value and corporate policies when the government imposes a binding quota on board composition?
- **Policy shock.** Norway required public limited firms to have at least 40% female directors. Before the law, women held only a small minority of board seats.
- **Core treatment intensity.** Firms with fewer women directors before the quota were farther from compliance and therefore faced a larger forced board-composition shock.
- **Main evidence.** Firms most constrained by the quota had negative announcement returns and lower subsequent Tobin's Q.
- **Mechanism.** The quota forced rapid director replacement. New women directors were younger and had less CEO experience than retained male directors, and affected firms changed leverage, acquisitions, and operating outcomes.
- **Interpretation.** The estimates capture the effect of a sudden binding board-composition constraint, not the timeless or universal effect of female directors.
- **Takeaway.** Boards matter for shareholder value and policy. Diversity mandates can rapidly change representation, but fast binding quotas can impose adjustment costs when the experienced candidate pool is limited.

2 Data and measurement

The sample covers Norwegian public limited companies with publicly traded shares from 2001–2009. The final Norwegian sample contains 1,230 firm-year observations for 248 firms. Board and CEO information is hand-collected from annual reports, the Register of Business Enterprises, tax records, and BoardEx. Accounting and market variables come from Compustat Global, CRSP for U.S. data, exchange-rate sources, and CPI adjustments.

2.1 Key variables

- **Percent female directors:** share of shareholder-elected directors who are women.
- **Tobin's Q:** total assets minus book equity plus market equity, divided by total assets.
- **Director human capital:** CEO experience, age, tenure, education, MBA status, and outside occupation.
- **Firm policies:** leverage, current debt/equity, cash/assets, capex/assets, acquisitions/assets, R&D/assets, and performance measures.
- **Avoidance outcomes:** whether firms delist from the public limited form or switch organizational status to avoid the quota.

Interpretation: The empirical object is not simply gender diversity. The paper measures a forced change in the feasible board set: firms had to move from their chosen pre-quota composition toward a legal 40% target in a short period.

3 Models and empirical specifications

3.1 Event-study returns

$$AR_i = \sum_{t \in \mathcal{W}} (r_{it} - r_{industry,t}).$$

Interpretation: The event study measures the immediate market-value response to the unexpected quota announcement. The main comparison is between firms with no women directors and firms with at least one woman director before the law.

3.2 Panel model for Tobin’s Q

$$Q_{it} = \alpha + \beta \text{PercentFemale}_{it} + \theta_i + \tau_t + \varepsilon_{it}.$$

Because board composition is endogenous, the paper instruments current female board representation with pre-quota female board representation interacted with year dummies:

$$\text{PercentFemale}_{it} = \sum_{k \neq 2003} \pi_k (\text{PercentFemale}_{i,2002} \times \mathbf{1}\{t = k\}) + \theta_i + \tau_t + \eta_{it}.$$

Interpretation: Firms farther from the quota before the law had to make larger changes after the mandate. The IV estimate is therefore a local effect of quota-induced changes in board composition among firms constrained by the law.

3.3 Mechanism and avoidance regressions

For board characteristics and firm policies:

$$Y_{it} = \alpha + \beta \widehat{\text{PercentFemale}}_{it} + \theta_i + \tau_t + \varepsilon_{it}.$$

For quota avoidance:

$$\text{Pr}(\text{Delist}_i = 1) = F(\alpha + \beta \text{PercentWomen}_{i,2002} + \gamma X_{i,2002}).$$

Interpretation: These tests ask whether the quota changed board human capital, firm size, leverage, acquisitions, performance, and the probability of leaving the regulated organizational form.

4 Identification and empirical design

The identification strategy has two complementary parts: an event-study design around the unexpected policy announcement and a long-run IV design that uses pre-quota board composition as predetermined exposure to the mandate.

4.1 Policy setting and treatment intensity

Norway’s law applied to public limited companies and required at least 40% female board representation. Treatment intensity is not defined by whether a firm is Norwegian; all covered firms are treated. Instead, treatment intensity is defined by how far the firm was from the quota before the law:

$$\text{DistanceToQuota}_i = 0.40 - \text{PercentFemale}_{i,pre}.$$

Firms with zero women directors had the largest required adjustment, while firms that already had women directors were closer to compliance.

Interpretation: This creates cross-sectional variation in the severity of the same legal mandate. The key empirical comparison is between firms that were far from compliance and firms that were already partly compliant.

4.2 Event-study identification

The February 2002 announcement was not widely anticipated. The event study compares abnormal returns across firms with different pre-announcement board composition:

$$AR_i = \alpha + \beta NoWomen_{i,pre} + \gamma X_i + u_i.$$

- AR_i is the short-window abnormal return.
- $NoWomen_{i,pre}$ marks firms with no women directors before the announcement.
- X_i includes controls such as size, industry, and governance characteristics in some specifications.

Interpretation: The event-study coefficient captures the market’s immediate assessment of the quota’s cost or benefit for firms that had to change their boards the most. Because the window is short, slow-moving omitted variables are less likely to explain the result.

4.3 IV identification for long-run valuation

The long-run design instruments $PercentFemale_{it}$ with 2002 female board representation interacted with post-quota year indicators:

$$Z_{it,k} = PercentFemale_{i,2002} \times \mathbf{1}\{t = k\}.$$

The first stage is strong because firms with fewer women in 2002 had to add more women to comply. The second stage estimates the impact of the quota-induced increase in female board representation on Tobin’s Q.

Interpretation: The instrument uses predetermined board composition, not contemporaneous board choices. This helps address the concern that firms voluntarily choosing more women directors differ in unobserved ways from other firms.

4.4 Reduced form, first stage, and IV logic

The IV design can be read as three linked objects:

1. **First stage:** pre-quota female representation predicts post-quota female representation.
2. **Reduced form:** pre-quota female representation predicts post-quota firm value.
3. **IV ratio:** the long-run value effect equals the reduced-form value effect scaled by the first-stage board-composition effect.

Interpretation: This structure is important because the IV coefficient is not a generic diversity coefficient. It is the effect of the board change forced by the law.

4.5 Main identifying assumptions

- **Relevance.** Firms with fewer women directors before the law must change board composition more after the quota.
- **Exclusion restriction.** Conditional on firm and year fixed effects, 2002 female representation affects later Tobin’s Q mainly through the mandated board-composition change.
- **Parallel trends.** Firms with and without pre-quota women would have followed similar valuation trends absent the quota.
- **No anticipation.** 2002 board composition was not chosen in anticipation of the later binding quota.
- **Stable comparison.** Sample selection through delisting should not fully drive the valuation results.

Interpretation: These assumptions are plausible because the policy was a sharp legal shock, but they are not automatic. The paper therefore combines announcement returns, first stages, reduced forms, placebo comparisons, and avoidance tests.

4.6 Why the estimates should not be read as the pure effect of women directors

The law changed multiple objects at once:

- the gender composition of boards;
- the age and experience distribution of directors;
- the turnover rate of boards;
- the supply-demand balance in the director labor market;
- firms' legal-form and delisting incentives.

Interpretation: The estimate is the causal effect of a rapid quota-induced board reorganization. It may include adjustment costs, scarcity of experienced candidates, disruption to board networks, and changes in group dynamics.

4.7 Threats and how the paper addresses them

- **Pre-existing trends.** Firms without women might already have been declining. The paper uses fixed effects, pre-period exposure, and placebo comparisons to reduce this concern.
- **Selection through delisting.** The most constrained firms may leave the public limited form. The delisting analysis is therefore a key part of the identification story, not only an auxiliary result.
- **Concurrent Norwegian shocks.** A Norway-wide shock would affect all Norwegian firms, but the results vary with pre-quota compliance status.
- **Voluntary diversity.** Voluntary appointment of women may be correlated with firm quality. The IV strategy avoids using voluntary contemporaneous board choices as the source of identification.
- **External validity.** The results apply to a rapid binding quota in a small director market; they need not imply that gradual diversity increases or larger candidate pools have the same effect.

Interpretation: The strongest interpretation is that a sudden legally mandated board-composition constraint imposed costs on firms farthest from compliance.

5 Tables and figures

5.1 Figure I: Percentage of Women Directors and CEOs of Norwegian Public Limited Firms; Table I: Boardroom Gender Quota Laws and Recommendations by Country

Read: Figure I shows female board representation rising above 40%, while female CEOs and chairpersons remain low. Table I places Norway's quota in international context and shows that Norway's rule was unusually early and binding.

Economic message: The mandate strongly changed board composition, not executive control.

on their boards of directors by July 2006, at the time women held only 9% of board seats. After voluntary compliance failed, the law became compulsory January 1, 2006, with a two-year transition period. Firms that did not comply by January 2008 would be forced to dissolve. Notices to comply were given to 77 delinquent firms in January 2008, and by April all public limited firms were in compliance with the law. Figure I presents the time series of this dramatic transformation in the composition of Norwegian boards of directors. Though more women were elected to boards,

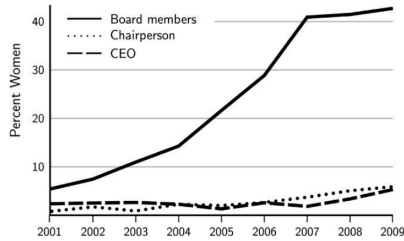


FIGURE I
Percentage of Women Directors and CEOs of Norwegian Public Limited Firms

(a) **Figure I. Percentage of women directors, chairpersons, and CEOs.** The board line jumps sharply because the mandate directly targets director seats. The CEO and chairperson lines remain low because the law does not directly require women in executive positions.

TABLE I
BOARDROOM GENDER QUOTA LAWS AND RECOMMENDATIONS BY COUNTRY

Country	Year quota passed	Mandatory quota	Quota compliance year	Gender equality in governance code	Average percent of women on boards in 2010
Norway	2003	40%	2008	2009	39
Spain	2007	40%	2015	2006	10
Iceland	2010	40%	2013		16
Finland	2010 ^a	1 Woman	2010	2010	26
France	2011	40%	2017	2010	12
Belgium	Pending ^b	33%	N/A	2009	10
Netherlands	Pending ^b	30%	2015	2010	15
Italy	Pending ^b	30%	2015		5
Sweden	Discussion ^d			2004	26
Germany	Discussion ^e			2009	13
United Kingdom	Discussion ^f			2010	9
Canada	Discussion ^g				14
Denmark				2008	18
Luxembourg				2009	4
Australia				2009	8
United States				2009	16
Austria				2010	9
Poland				2010	12

Notes: Pending indicates that a law has passed at least one stage of the legislative process. Discussion indicates that media sources cite politicians debating a quota. Gender equality in governance code indicates the year in which the country's code of good governance recommended gender equality. Average percent of women on the board of directors in 2010 for European countries is taken from the European Commission Database on Women & Men in Decision Making. For non-European countries the data are from Catalyst. Data for Year quota passed is from European Commission (2011), unless otherwise noted: a: Legislative Document No. 5-0607 of the Belgian Senate; b: Vijnshar (2011); c: PIRC (2011); d: Sweden had discussions in 2003 (Lundvall 2003); e: Fox (2011); f: 2011 Lord Davies Report; g: Rott (2010). Note that Finland's quota is in the governance code and requires firms to have at least one woman director or to explain why they do not.

(b) **Table I. Boardroom gender quota laws and recommendations by country.** Norway stands out because the quota is binding, early, and backed by enforcement; many other countries use recommendations or weaker targets.

Detailed interpretation: The figure is the first-stage visual evidence: the policy did what it was designed to do. The fact that CEO and chairperson representation barely changes is important because the treatment is a board-composition shock rather than a general female-leadership shock. The table clarifies external validity: Norway is not just another governance reform; it is a strict mandate with stronger enforcement than most international governance recommendations.

5.2 Table II: Firm and Board of Directors Summary Statistics by Year

Read: Female board share rises from 5.42% in 2001 to 42.62% in 2009. Board size remains roughly stable, while CEO experience and some board human-capital measures fall.

Economic message: Firms largely replaced male directors rather than simply expanding boards.

TABLE II
FIRM AND BOARD OF DIRECTORS SUMMARY STATISTICS BY YEAR

Panel A. Board characteristics	2001	2002	2003	2004	2005	2006	2007	2008	2009
Number of members	5.54	5.53	5.39	5.32	5.38	5.60	5.63	5.57	5.29
Female (%)	5.42	7.47	10.97	14.29	21.64	29.75	40.80	41.39	42.62
CEO experience (%)	73.62	64.86	60.21	60.01	61.52	59.03	56.09	55.55	56.44
MBA (%)	23.45	23.69	25.71	25.65	26.39	27.88	25.82	24.42	21.63
Higher education (%)	25.38	26.15	26.14	26.42	26.69	26.13	26.60	27.42	26.67
Age	56.47	51.25	51.47	51.79	50.98	50.75	50.93	51.34	52.28
Tenure (years) ^a	4.33	4.03	3.46	3.40	3.17	2.92	2.19	2.23	2.58
Inside (%)	2.22	2.21	2.67	4.96	2.72	1.54	2.33	1.65	1.60
Ret. dir. (years)	7.87	6.15	6.06	6.33	6.17	6.02	6.07	6.24	7.07
Ret. dir. (tenure) ^a	2.25	1.60	1.36	1.36	1.50	1.75	1.82	1.76	1.76
Percent returned from prior year	76.22	80.23	82.36	74.66	80.64	71.41	76.33	76.30	
Norwegian (%)	89.24	92.75	90.46	90.17	91.15	91.72	91.46	89.80	89.09
Board size as other on board (%)	4.05	5.19	4.09	4.83	5.36	2.89	3.33	3.84	5.16
Board of CEO positions/person	2.44	2.28	2.52	2.82	2.80	2.34	2.71	3.11	2.84
Position/board size	1.98	2.03	2.11	2.25	2.44	2.86	3.26	3.71	2.53

(a) **Table II, Panel A. Board characteristics by year.** Shows the rapid increase in female director share and changes in age, tenure, and CEO experience.

TABLE II
(CONTINUED)

Panel B. Outside occupation of shareholder elected directors (%)	2001	2002	2003	2004	2005	2006	2007	2008	2009
Non-executive manager	5.38	5.05	4.40	4.91	6.96	4.43	3.76	6.71	6.77
CPD	3.87	3.89	4.40	3.97	5.40	5.85	6.04	6.14	5.94
Partnership/owner	17.38	17.07	17.13	16.44	18.51	17.55	17.87	16.84	16.39
Accountant	0.00	0.00	0.00	0.79	0.33	0.00	0.38	0.00	0.00
Other	9.47	9.97	9.70	9.71	9.09	4.48	9.84	9.86	9.53

(b) **Table II, Panel B. Outside occupations by year.** Shows how the occupational mix of directors changes as the quota binds.

TABLE II
(CONTINUED)

Panel C. Firm characteristics	2001	2002	2003	2004	2005	2006	2007	2008	2009
Turnover (%)	1.44	1.21	1.54	1.67	1.88	1.58	1.71	1.11	1.35
Log(assets)	4.48	4.38	4.50	5.03	5.14	5.44	5.71	5.57	5.62
Log(revenue)	4.46	4.62	4.70	4.67	4.81	5.06	5.13	5.07	5.06
Log(profit)	56.97	56.44	54.04	51.96	51.85	53.38	54.03	57.24	57.27
Current debt/equity (%)	36.26	27.28	11.97	12.51	21.36	22.49	19.09	16.71	21.07
Current debt/equity (%)	13.13	14.05	16.01	16.75	17.15	19.61	17.08	14.09	14.39
Capitalization (%)	7.26	5.26	4.94	4.97	4.22	7.41	6.53	6.43	6.20
Acquisition/losses (%)	0.58	0.51	0.65	1.34	1.27	2.19	2.86	1.15	0.40
RD/Revenue (%)	0.96	0.86	0.97	1.15	0.96	1.48	2.05	2.34	2.15
Observations	127	119	113	131	151	155	163	148	115

Notes: This table presents averages of firm characteristics and average of average level of director characteristics across firms for Norwegian companies that were listed on the Oslo Stock Exchange. Other variables: RD=research and development; RD/Revenue is an average of RD/Revenue ratio and RD/Revenue ratio at the end of 2009. *T-tests statistics. Average means and standard deviation (brackets) are from both mixed regressions and the control regressions using different specifications in the model. Because we are able to calculate means for these variables in the mixed regressions and because there is no control regressions, the means variables are used in the regressions in the mixed regressions that they are in the mixed regressions.

(c) **Table II, Panel C. Firm characteristics by year.** Tracks firm-level outcomes during the quota transition period.

Detailed interpretation: The most important detail is stable board size. If boards simply expanded, the law might have added diversity without displacing incumbents. Instead, stable board size implies substitution: firms had to replace existing male directors with new women directors. This makes changes in director experience and board networks central to the mechanism. Panel C should be read cautiously because

firm characteristics also move with macroeconomic and sample-selection forces; it is descriptive rather than causal.

5.3 Table III: Stock Returns at the Initial Announcement of the Boardroom Gender Quota

Read: Firms with no women directors experience negative announcement returns, while firms with at least one woman director have returns close to zero. Cross-country tests show that the negative reaction is concentrated in Norwegian firms exposed to the mandate.

Economic message: Investors priced the quota as costly for firms farthest from compliance.

TABLE III
STOCK RETURNS AT THE INITIAL ANNOUNCEMENT OF THE BOARDROOM GENDER QUOTA

Panel A. Abnormal announcement returns (%) of Norwegian firms			
All firms	No women directors	Women directors > 0	Difference
(1)	(2)	(3)	(2)-(3)
Mean	-2.573*** (0.001)	-2.547*** (0.001)	-0.024 (0.977)
Median	-1.804*** (0.017)	-2.521*** (0.010)	-1.593* (0.045)
Observations	94	68	26

(a) **Table III, Panel A. Norwegian announcement returns.** Firms with no women directors have the largest negative reaction, matching the predicted compliance burden.

TABLE III
(CONTINUED)

	Norwegian and U.S. firms				Scandinavian firms	
	(1)	(2)	(3)	(4)	(5)	(6)
Norwegian dummy	-4.347*** (1.468)	-4.146*** (1.404)	-3.803** (1.571)	-3.574** (1.525)	-4.026** (1.671)	-3.773** (1.644)
Women directors > 0	0.046 (0.320)		-0.214 (0.381)		-1.536 (1.041)	
Percentage women directors		0.594 (1.928)		-0.390 (2.319)		-9.457 (6.344)
Norwegian $\times$ women directors > 0	3.477** (1.648)		3.252* (1.658)		4.775** (1.897)	
Norwegian $\times$ percentage women directors		14.342* (7.589)		12.517 (7.861)		23.724** (10.746)
Board size			-0.052 (0.068)	-0.060 (0.065)	0.246 (0.314)	0.287 (0.323)
Log(assets)			0.362** (0.149)	0.350** (0.150)	0.021 (0.305)	-0.039 (0.321)
Constant	0.799*** (0.271)	0.776*** (0.265)	-1.206 (1.093)	-1.137 (1.088)	-0.920 (1.687)	-0.954 (1.715)
Adjusted R^2	0.025	0.024	0.030	0.029	0.056	0.053
Observations	1,252	1,252	1,224	1,224	205	205

Notes: Abnormal returns are the sum of industry-adjusted returns (using U.S. industry returns) over the five days surrounding the date of the first announcement of the gender quota on February 22, 2002. *Women directors > 0* is a dummy variable equal to 1 if the firm has at least 1 woman director. *Percentage women directors* is the percentage of the shareholder-elected board members that are women. *Board size* is the number of board members. Board data are taken from the 2001 annual reports. Regressions are cross-section OLS regressions on abnormal returns of firms located in the countries indicated at the top of each column. Scandinavian firms include firms in Norway, Denmark, Finland, and Sweden. Statistical significance is reported as p-values in Panel A (t-tests for the mean and sign and rank-sum tests for the medians) and robust standard errors clustered by industry in Panel B. * Significant at 10%; ** 5%; *** 1%.

(b) **Table III, Panel B. Cross-country announcement-return regressions.** The reaction is compared against firms outside Norway to separate the quota news from general market movements.

Detailed interpretation: This table is the cleanest short-run valuation evidence because the event window is narrow and the announcement was unexpected. The sorting by pre-existing women directors is crucial: firms with no women are the most constrained and therefore have the strongest negative reaction. The table does not prove the precise mechanism, but it shows that investors immediately expected the constraint to reduce value for the firms facing the largest adjustment.

5.4 Table IV: Effects of Board Member Gender Quotas on Tobin's Q

Read: The IV coefficient for Norway is negative and significant. The paper interprets a forced 10 percentage point increase in women directors as reducing Tobin's Q by about 12.4% relative to the average.

Economic message: The table provides the main long-run value result.

TABLE IV
EFFECTS OF BOARD MEMBER GENDER QUOTAS ON TOBIN'S Q

	Norway	Placebo tests	
		Denmark, Finland, and Sweden	U.S.
	(1)	(2)	(3)
Panel A. Instrumental variables regressions: dependent variable = industry-adjusted Q			
Percent women directors _{t}	-1.938*** (0.586)	-3.635 (2.352)	-0.264 (0.465)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
F -statistic	29.790	20.240	2.690
Observations	603	634	2,706
Panel B. Reduced-form regressions: dependent variable = industry-adjusted Q			
2004 dummy $\times$ percent of women in 2002	-0.340 (0.228)	-0.008 (0.397)	-0.027 (0.078)
2005 dummy $\times$ percent of women in 2002	-0.261 (0.416)	0.292 (0.556)	0.077 (0.096)
2006 dummy $\times$ percent of women in 2002	0.535 (0.492)	1.443** (0.737)	0.080 (0.092)
2007 dummy $\times$ percent of women in 2002	1.124** (0.512)	1.112 (0.926)	0.028 (0.112)
2008 dummy $\times$ percent of women in 2002	1.352*** (0.492)	0.902 (0.717)	-0.085 (0.129)

Table IV, part 1. IV and reduced-form effects on Tobin's Q . The second-stage estimates translate quota-induced board changes into valuation effects.

	Placebo tests		
	Norway	Denmark, Finland, and Sweden	U.S.
	(1)	(2)	(3)
2009 dummy $\times$ percent of women in 2002	1.980* (0.555)	1.246* (0.690)	-0.053 (0.148)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
F -statistic	15.840	18.130	2.430
Observations	605	721	2,999
Panel C. First-stage regressions: dependent variable = percent women directors _{t}			
2004 dummy	0.047*** (0.012)	0.041*** (0.009)	0.010*** (0.003)
2005 dummy	0.128*** (0.017)	0.059*** (0.011)	0.024*** (0.004)
2006 dummy	0.212*** (0.017)	0.095*** (0.013)	0.037*** (0.005)
2007 dummy	0.354*** (0.013)	0.092*** (0.014)	0.041*** (0.005)
2008 dummy	0.368*** (0.014)	0.110*** (0.015)	0.049*** (0.005)

	Placebo tests		
	Norway	Denmark, Finland, and Sweden	U.S.
	(1)	(2)	(3)
2009 dummy	0.368*** (0.014)	0.114*** (0.033)	0.055*** (0.006)
2004 dummy $\times$ percent of women in 2002	-0.117 (0.073)	-0.177** (0.077)	-0.050** (0.025)
2005 dummy $\times$ percent of women in 2002	-0.274*** (0.091)	-0.188 (0.119)	-0.139*** (0.036)
2006 dummy $\times$ percent of women in 2002	-0.419*** (0.087)	-0.412*** (0.116)	-0.190*** (0.046)
2007 dummy $\times$ percent of women in 2002	-0.793*** (0.089)	-0.286** (0.115)	-0.195*** (0.047)
2008 dummy $\times$ percent of women in 2002	-0.807*** (0.082)	-0.344*** (0.116)	-0.250*** (0.044)
2009 dummy $\times$ percent of women in 2002	-0.799*** (0.091)	-0.546* (0.293)	-0.277*** (0.051)
Firm fixed effects	Yes	Yes	Yes
F -statistic	19.670	6.150	8.400
Observations	603	634	2,706

Notes: Data are yearly observations from 2003 to 2009. Percent women directors _{t} is a predicted variable estimated in the first stage regressions. Percent of women in 2002 reports the percentage of shareholder-elected directors that were women as reported in the firm's 2002 annual report. Year 2001 variables are omitted. Standard errors are clustered by firm and are reported in parentheses. * Significant at 10%; ** 5%; *** 1%.

(a) Table IV, part 2. Reduced-form estimates. Shows how predetermined compliance status predicts post-quota valuation outcomes.

(b) Table IV, part 3. First-stage estimates and notes. Documents that pre-quota board composition strongly predicts quota-induced changes.

Detailed interpretation: Table IV should be read as a LATE-style estimate for firms moved by the quota. The first stage is a compliance-stage result: firms far from the legal target had to add more women. The reduced form is a valuation result: the same predetermined exposure predicts lower Q after the law. The IV estimate combines these two pieces. The coefficient is not the effect of one voluntary female director; it is the effect of a forced, rapid board-composition adjustment.

5.5 Table V: Characteristics of New, Retained, and Exiting Directors by Gender

Read: New women directors have much less CEO experience than retained male directors and are younger. The entrant pool differs from both retained men and exiting men.

Economic message: The table supports the mechanism that the quota changed board human capital.

	New		Retained		Exiting		Differences			
	Women	Men	Women	Men	Women	Men	(1)-(3)	(1)-(5)	(1)-(6)	(4)-(6)
Panel A: Demographics										
CEO Exp. (%)	31.18	63.86	42.97	69.41	36.02	65.17	-39.24***	-34.00***	4.24*	
							(-15.52)*	(-11.284)	(1.930)	
MBA (%)	26.09	35.82	28.06	24.31	25.82	24.81	1.78	1.27	-0.50	
							(0.720)	(0.446)	(-0.241)	
Higher educ. (%)	33.09	39.46	35.41	24.97	30.06	23.27	6.13***	9.83***	3.70	
							(3.263)	(5.120)	(0.852)	
Age	45.84	50.37	48.24	54.08	47.58	52.94	-8.24***	-7.10***	1.14**	
							(-33.613)	(-16.079)	(3.252)	
Tenure	0.00	0.00	2.19	3.94	1.37	2.54	-3.94***	-2.54***	1.40**	
							(-38.963)	(-16.000)	(7.744)	
Insider (%)	1.42	4.36	0.88	5.33	1.50	7.05	-2.91***	-5.62***	-1.72	
							(-4.322)	(-8.176)	(1.250)	
Norwegian (%)	87.55	85.53	90.28	90.04	88.28	87.88	-2.49	-0.33	2.16	
							(-1.325)	(-0.135)	(1.490)	
Same last name	3.33	4.62	4.06	6.98	2.81	4.35	-0.65***	-1.01	2.64***	
							(-4.325)	(-1.389)	(5.599)	

	New		Retained		Exiting		Differences			
	Women	Men	Women	Men	Women	Men	(1)-(3)	(1)-(5)	(1)-(6)	(4)-(6)
Panel B: Primary outside occupation (%)										
VP	13.49	6.81	10.72	5.03	14.59	4.96	7.60***	6.51***	0.87	
							(3.603)	(3.073)	(0.494)	
Consultant	14.13	9.81	6.90	9.43	9.03	9.66	4.74**	4.82*	-0.3	
							(2.143)	(1.759)	(-0.120)	
Board member	9.03	16.09	15.71	20.36	11.38	19.32	-26.47***	-9.39***	1.61	
							(-5.081)	(-3.485)	(0.457)	
Professor	2.13	1.09	3.23	1.53	3.01	1.21	6.90	0.83	6.81	
							(0.877)	(0.792)	(0.040)	
CEO	16.21	26.24	21.59	27.14	19.35	25.85	-20.67***	-16.64***	1.39	
							(-4.280)	(-3.035)	(0.510)	
Attorney	5.32	2.18	4.70	3.30	4.51	3.39	2.02	1.92	-0.10	
							(1.422)	(1.182)	(-0.094)	

	New		Retained		Exiting		Differences			
	Women	Men	Women	Men	Women	Men	(1)-(3)	(1)-(5)	(1)-(6)	(4)-(6)
Nonexec. mgt.	14.54	4.62	9.25	2.34	14.39	4.79	12.06***	9.84***	-2.10*	
							(5.606)	(4.160)	(-1.879)	
CFO	7.09	4.90	6.17	3.23	9.02	3.13	5.86***	3.96**	0.10	
							(2.419)	(2.234)	(0.101)	
Partner/principal	9.93	20.36	13.22	19.78	7.52	18.54	-9.36***	-8.61***	1.29	
							(-4.915)	(-3.225)	(0.590)	
Accountant	0.35	0.00	0.15	0.13	0.00	0.00	0.23	0.35	0.13	
							(0.623)	(1.000)	(1.413)	
Other	5.32	0.82	7.49	1.92	6.02	2.08	3.86***	3.23**	-0.57	
							(2.764)	(2.117)	(-0.714)	
Observations	800	805	1,382	3,733	285	1,150				

Notes: Averages of personal characteristics of shareholder-elected directors in Norwegian firms from 2001 to 2009. New refers to directors that are new to a board. Retained are directors that were on the board in the prior year. Exiting are directors that were on the board in the prior year, but not in the following year. Observations are gender-year. Statistical significance is reported by calculating a p-value from two-sample tests comparing sample variances. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and * respectively and added to the right of the coefficient.

(a) Table V, Panel A. Demographics. New women directors are younger and have less CEO experience.

(b) Table V, Panel B. Outside occupations, part 1. Shows changes in director backgrounds and professional categories.

(c) Table V, Panel B. Outside occupations, part 2. Completes the occupational-composition comparison.

Detailed interpretation: This is the main mechanism table. The binding quota created simultaneous demand for women directors, so firms had to search in a thinner director labor market. The decline in CEO experience is economically important because boards advise management, monitor executives, and approve acquisitions and financing. The table also implies that the valuation result may reflect constrained director supply and rapid turnover, not gender per se.

5.6 Table VI: Instrumental Variables Estimates of the Effect of the Gender Quota on Firm Characteristics

Read: The quota is associated with changes in firm size, board experience, leverage, acquisitions, cash, and performance. Some outcomes move in directions consistent with less experienced boards or changed board preferences.

Economic message: Mechanism results show that the quota changed corporate policies as well as board demographics.

TABLE VI
INSTRUMENTAL VARIABLES ESTIMATES OF THE EFFECT OF THE GENDER QUOTA ON FIRM CHARACTERISTICS

Panel A. Firm size and corporate leadership												
	Log (assets)	Log (empl.)	Board size	CEO exper.	Age	MBA	Higher educ.	New CEO	Woman CEO	CEO comp.		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)		
Percent women	2.00** (0.81)	2.10** (0.86)	0.49 (1.07)	-59.80** (23.70)	-7.115* (3.65)	-9.72 (33.20)	34.30 (22.10)	-25.70 (37.70)	25.90 (21.60)	2.33 (1.87)		
Observations	639	479	640	605	640	578	578	637	640	325		
Panel B. Financial and investment policies												
	Leverage		Current debt/equity		Cash/ assets		Capex/ assets		Acquisitions/ assets		R&D/ assets	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Percent women	0.32** (0.15)	0.24 (0.23)	0.53* (0.30)	1.37 (1.08)	-0.10 (0.10)	-0.27** (0.11)	-0.04 (0.05)	-0.08 (0.06)	0.07*** (0.03)	0.23*** (0.07)	0.01 (0.02)	0.04 (0.03)
Intl. accounting std.	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Observations	642	422	638	418	633	421	633	415	627	412	642	422

(a) Table VI, Panels A and B. Firm size, leadership, financial policies, and investment policies. The estimates link quota-induced board changes to policy variables.

TABLE VI
(CONTINUED)

Panel C. Performance and costs									
	M&A returns	Daily returns	ROA		Asset turnover		SGA/ sales		COGS sales
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Percent women	0.01 (0.03)	-0.45** (0.22)	-0.07 (0.11)	-0.21 (0.17)	-0.34 (0.48)	-1.24* (0.64)	0.05 (0.11)	0.17* (0.10)	1.21 (1.40)
Intl. accounting std.	No	No	No	Yes	No	Yes	No	Yes	No
Observations	635	550	610	414	642	422	575	404	577

Notes: Dependent variables are listed at the top of each column. Data include Norwegian firm-years from 2003 to 2009. Percent women is a predicted variable, estimated in a first stage regression using the percent of women board members in 2002 interacted with year dummies. First stage estimates are not reported for brevity. All accounting variables are industry-adjusted, using U.S. industry values. Definitions of all variables are available in Appendix A. CEO exper., MBA, Higher educ., indicate the fraction of shareholder-elected directors that have prior experience as a CEO, have an MBA, or have another postsecondary degree. New CEO and Woman CEO are dummy variables to indicate if a new or woman CEO was appointed. CEO comp. is total compensation paid to the CEO from the Boardstar database. M&A Return is the yearly sum of five-day abnormal announcement returns per firm; a zero is recorded for firm-years with no acquisitions. All regressions include year and firm fixed effects. Intl. accounting std. indicates whether firm-years included in the sample are restricted to accounting disclosures that follow international accounting standards. Standard errors are clustered by firm and reported in parentheses. * Significant at 10%; ** 5%; *** 1%.

(b) Table VI, Panel C. Performance and costs. Shows effects on profitability and cost-related outcomes.

Detailed interpretation: Table VI is not a single-mechanism proof. It is a pattern-of-outcomes table. If the quota only changed demographic labels without changing board function, we would not expect systematic changes in acquisitions, leverage, cash, or performance. The pattern suggests real governance consequences. At the same time, the table cannot fully distinguish whether the channel is lower CEO experience, younger directors, disruption from turnover, altered preferences, or changed board deliberation.

5.7 Figure II: Number of Firms by Organizational Status in Norway; Table VII: Likelihood of Delisting by Board Gender

Read: Public limited firms decline while private limited firms rise. Firms with fewer women before the quota are more likely to delist or avoid the quota.

Economic message: The quota imposed costs large enough that some firms changed legal status rather than comply.

Since 2001, there has been a steady increase in the number of private limited firms than existed in 2001. These changes occurred at a time when employment was increasing, with the exception of 2009. This indicates that while the economy was growing, firms were more likely to choose to organize as a private limited firm rather than a public limited firm starting in the years after the announcement of the gender quota law, which only affects public limited firms. These data do not allow us to identify firms that

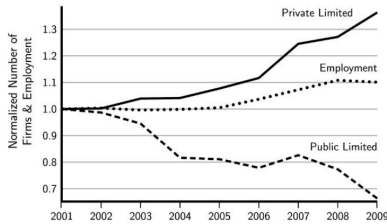


FIGURE II
Number of Firms by Organizational Status in Norway

(a) Figure II. Number of firms by organizational status. Shows movement away from the regulated public limited form during the quota period.

TABLE VII
LIKELIHOOD OF DELISTING BY BOARD GENDER

	Dependent Variable: Delist for any reason			Dependent Variable: Avoid quota through delisting		
	(1)	(2)	(3)	(4)	(5)	(6)
Percent women directors	-6.604*** (1.724)	-8.959*** (2.151)	-9.763*** (1.801)	-7.919*** (2.725)	-9.143*** (3.376)	-10.460*** (3.998)
Board size	-0.093 (0.146)	-0.194 (0.176)	-0.250 (0.212)	-0.306* (0.179)	-0.374 (0.244)	-0.472* (0.256)
Board age	-0.018 (0.022)	-0.086** (0.039)	-0.082* (0.045)	-0.025 (0.027)	-0.070 (0.047)	-0.073 (0.048)
Board CEO experience	-1.304** (0.659)	-1.511*** (0.534)	-2.159*** (0.758)	-0.809*** (0.257)	-0.903*** (0.313)	-1.834*** (0.563)
Log(assets)		0.378** (0.173)	0.488*** (0.131)		0.247 (0.295)	0.439 (0.358)
R&D/assets		4.421 (7.974)	5.912 (12.681)		-0.803 (6.685)	-2.401 (9.027)
Industry:			0.419 (0.257)		-0.115 (0.111)	
industrials						
consumer discretionary			-0.109 (0.245)		-0.699*** (0.252)	

(b) Table VII, part 1. Likelihood of delisting by board gender. Firms with fewer pre-quota women are more likely to avoid the mandate.

TABLE VII
(CONTINUED)

	Dependent Variable: Delist for any reason			Dependent Variable: Avoid quota through delisting		
	(1)	(2)	(3)	(4)	(5)	(6)
Industry:			0.871***			1.284***
consumer staples			(0.258)			(0.370)
Industry:			-1.141***			-1.852***
health care			(0.138)			(0.419)
Industry:			-0.466			0.590
information technology			(1.863)			(1.377)
Industry:			0.858***			0.761
utilities			(0.281)			(0.559)
Constant			4.332***			4.753**
			(1.550)			(1.961)
Observations	95	95	91	95	95	91
Pseudo R^2	0.112	0.144	0.191	0.131	0.144	0.215

Notes: Logit regression estimates where the dependent variable equals 1 if the firm delisted from 2003 to 2009 and 0 otherwise in columns (1)–(3), or equals 1 if the firm avoided the quota through delisting (acquisition by a private/foreign firm, went private, or changed location) or 0 otherwise (didn't delist, went bankrupt, or no reason can be determined). Variables are based on 2002 data only. Omitted industry sector is energy. Standard errors in parentheses are robust to industry sector clusters. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *. All variables are defined in Appendix A.

Table VII, part 2. Industry controls and notes. Shows robustness of the quota-avoidance relationship.

Detailed interpretation: The avoidance evidence matters for both economics and identification. Economically, it reveals that compliance costs were not trivial. Empirically, it warns that post-quota valuation estimates may be selected because the firms most harmed by the quota may leave the sample. This means the main value estimates may understate the total cost of the mandate for all originally covered firms.

6 Short summary

- Norway's quota raised female board representation above 40%.
- Firms farthest from compliance had negative announcement returns.
- IV estimates indicate a large negative effect on Tobin's Q for quota-induced board changes.
- New women directors were younger and had less CEO experience than retained men.
- The quota affected leverage, acquisitions, cash, performance, and delisting.
- The evidence supports the claim that boards affect firm value and policy.
- The estimates should be interpreted as the effect of a rapid binding quota, not the pure effect of female directors.

7 Conclusion

7.1 Core conclusion

Ahern and Dittmar show that Norway's board gender quota had economically large effects on board composition, firm valuation, corporate policy, and organizational form. The quota achieved its representation objective quickly, but firms farthest from compliance experienced negative valuation effects.

7.2 Economic mechanism

The most plausible mechanism is that the quota forced a rapid reorganization of boards in a market with a limited pool of experienced women directors. The evidence on new director age and CEO experience supports this supply-constrained interpretation. Because boards provide monitoring, advice, approval of major investments, and financing oversight, a change in director human capital can affect firm policy and value.

7.3 Identification takeaway

The empirical design is compelling because it combines two sources of evidence. The event study captures a short-run market reaction to an unexpected policy announcement, while the IV design uses predetermined pre-quota board composition to measure long-run effects. The key identifying comparison is not women versus men in general; it is firms forced to change their boards a lot versus firms forced to change less.

7.4 Policy interpretation

The paper does not imply that diversity is bad. It shows that rapid binding mandates can impose adjustment costs when the relevant candidate pool is constrained. A slower transition, investments in director pipeline development, or broader eligibility pools could lead to different outcomes. Thus, the paper is best read as evidence on the design and timing of quotas, not as a universal statement about women directors.

7.5 Broader takeaway

The paper provides clean evidence that corporate boards matter. It also highlights a general lesson for regulation: policy can achieve a compositional target while changing the human capital, incentives, and selection of regulated firms. The full economic effect of a mandate includes both direct compliance and indirect responses such as delisting, legal-form switching, and altered corporate policies.